

Завдання 1

```
1 d = 23
2 c = 6
3
4 print(f"Numbers from {d} to 31:", end=" ")
5
6 count = 0
7 total_sum = 0
8 product = 1
9 even_count = 0
10 odd_count = 0
11
12 for num in range(d, 32):
13     print(num, end=" ")
14     count += 1
15     total_sum += num
16     product *= num
17
18     if num % 2 == 0:
19         even_count += 1
20     else:
21         odd_count += 1
22
23 print()
24 average = total_sum / count
25
26 print(f"Count: {count}")
27 print(f"Sum: {total_sum}")
28 print(f"Product: {product}")
29
30 print(f"Average: {average:.2f}")
31 print(f"Even: {even_count}, odd: {odd_count}")
32
33 #while version
34 count_w = 0
35 total_sum_w = 0
36 product_w = 1
37 even_count_w = 0
38 odd_count_w = 0
39
40 current = d
41 while current <= 31:
42     print(current, end=" ")
43     count_w += 1
44     total_sum_w += current
45     product_w *= current
46
47     if current % 2 == 0:
48         even_count_w += 1
49     else:
50         odd_count_w += 1
51
52     current += 1
53
54 print()
55 average_w = total_sum_w / count_w
56
57 print(f"Count: {count_w}")
58 print(f"Sum: {total_sum_w}")
59 print(f"Product: {product_w}")
60 print(f"Average: {average_w:.2f}")
61 print(f"Even: {even_count_w}, odd: {odd_count_w}")
62
63 print("Countdown:", end=" ")
64 for i in range(c, 0, -1):
65     print(i, end=" ")
66 print()
```

```
Numbers from 23 to 31: 23 24 25 26 27 28 29 30 31
Count: 9
Sum: 243
Product: 7315688016000
Average: 27.00
Even: 4, odd: 5
Numbers from 23 to 31: 23 24 25 26 27 28 29 30 31
Count: 9
Sum: 243
Product: 7315688016000
Average: 27.00
Even: 4, odd: 5
Countdown: 6 5 4 3 2 1
PS C:\Users\Windows 11 Pro\OneDrive\Desktop\Програмування\practik_4>
```

Завдання 2

```
1 print("Karina Batura, IT-31")
2
3 num = int(input("Enter number: "))
4
5 n = num
6
7 if n < 0:
8     n = -n
9
10 count = 0
11 total = 0
12 rev = 0
13
14 if n == 0:
15     count = 1
16     total = 0
17     mx = 0
18     mn = 0
19     rev = 0
20 else:
21     mx = 0
22     mn = 9
23
24     while n > 0:
25         digit = n % 10
26
27         count += 1
28         total += digit
29         rev = rev * 10 + digit
30
31         if digit > mx:
32             mx = digit
33         if digit < mn:
34             mn = digit
35
36         n //= 10
37
38 print(f"Digits: {count}")
39 print(f"Sum: {total}")
40 print(f"Max: {mx}, min: {mn}")
41 print(f"Reversed: {rev}")
```

```
...
Enter number: 23032009
Digits: 8
Sum: 19
Max: 9, min: 0
Reversed: 90023032
```

Завдання 3

```
task3.py > ...
1  print("Karina Batura, IT-31")
2
3  name = "Karina"
4  surname = "Batura"
5
6  f_name = name + surname
7
8  vowels = 0
9  consonants = 0
10
11  for char in f_name:
12      c = char.lower()
13
14      if c in "aeiouy":
15          vowels += 1
16      elif c.isalpha():
17          consonants += 1
18
19  total = vowels + consonants
20
21  print(f"{name} {surname}")
22  print(f"Vowels: {vowels}, consonants: {consonants}")
23  print(f"Total letters: {total}")
```

```
Karina Batura, IT-31
Karina Batura
Vowels: 6, consonants: 6
Total letters: 12
```

Завдання 4

```
print("Karina Batura, IT-31")

attempts = 0

while True:
    score = int(input("Enter your score (0-100): "))
    attempts += 1

    if score < 0:
        print("Score cannot be negative")
    elif score > 100:
        print("Score is too big, maximum is 100")
    else:
        break

print(f"Accepted after {attempts} attempts")

if score >= 90:
    grade = "A"
elif score >= 82:
    grade = "B"
elif score >= 74:
    grade = "C"
elif score >= 64:
    grade = "D"
elif score >= 60:
    grade = "E"
else:
    grade = "F"

print(f"Grade: {grade}")
```

```
Karina Batura, IT-31
Enter your score (0-100): -3
Score cannot be negative
Enter your score (0-100): 120
Score is too big, maximum is 100
Enter your score (0-100): 67
Accepted after 3 attempts
Grade: D
```

Завдання 5

```
1 print("Karina Batura, IT-31")
2
3 d = 23
4 c = 6
5 n = d * c
6
7 print(f"n = {d} * {c} = {n}")
8
9 div_count = 0
10 div_sum = 0
11
12 print("Divisors:", end=" ")
13 for i in range(1, n + 1):
14     if n % i == 0:
15         print(i, end=" ")
16         div_count += 1
17         div_sum += i
18 print()
19
20 print(f"count: {div_count}, sum: {div_sum}")
21
22 if n < 2:
23     print(f"{n} is not prime")
24 else:
25     for i in range(2, n):
26         if n % i == 0:
27             print(f"{n} is not prime")
28             break
29     else:
30         print(f"{n} is prime")
31
32 primes_count = 0
33
34 print(f"Primes up to {n}:", end=" ")
35 for num in range(2, n + 1):
36     for i in range(2, num):
37         if num % i == 0:
38             break
39     else:
40         print(num, end=" ")
41         primes_count += 1
42 print()
43
44 print(f"Primes count: {primes_count}")
```

```
Karina Batura, IT-31  
n = 23 * 6 = 138  
Divisors: 1 2 3 6 23 46 69 138  
count: 8, sum: 288  
138 is not prime  
Primes up to 138: 2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83 89 97 101 103 107 109 113 127 131 137  
Primes count: 33  
PS C:\Users\Windows 11 Pro\OneDrive\Desktop\Програмування\practik_4>
```